

2.3 PreQuiz: Motion and Kinematics

1. Round each value to three significant figures.

(a) 0.005283 m

(c) 8.0074 s

(b) 47.652 kg

(d) 15,340 N

2. Write each number in proper scientific notation ($1 \leq a < 10$).

(a) 0.000628

(c) 304,000,000

(b) 72,500

(d) 0.00915

3. Perform each operation and give the answer to 3 significant figures.

(a) $(4.8 \times 10^2) + (5.2 \times 10^3)$

(c) $\frac{8.4 \times 10^6}{2.1 \times 10^3}$

(b) $(7.20 \times 10^{-3}) \times (4.0 \times 10^4)$

4. Convert each value. Show one line of work using unit factors.

(a) 4.75 m to cm

(c) 0.450 kg to g

(b) 2.80 km to m

(d) 6.30×10^4 g to kg

5. Convert each speed.

(a) 72.0 km/h to m/s

(b) 8.50 m/s to km/h

6. A student walks along a straight hallway (forward is +). Start $x_0 = 3.0$ m.

(a) She walks to $x = 11.0$ m. Find displacement Δx .

(b) From $x = 11.0$ m she walks back to $x = 5.0$ m. Find Δx .

(c) Find total displacement from start to finish.

(d) Find total distance traveled.

7. A cart moves from $x_1 = 2.0$ m at $t_1 = 0$ s to $x_2 = 8.6$ m at $t_2 = 4.0$ s.

(a) Find displacement.

(b) Find average velocity v_{avg} .

8. One-dimensional vector additions: find the total displacement.

(a) $\Delta x_1 = +18$ m, $\Delta x_2 = -7$ m

(b) $+2.30$ km, -0.80 km, $+1.50$ km

Kinematics Equations

Formulas: $v = v_0 + at$ $d = v_0t + \frac{1}{2}at^2$ $v^2 = v_0^2 + 2ad$ $d = \frac{1}{2}(v_0 + v)t$

9. How far can a cyclist travel in 2.5 h along a straight road if her average velocity is 18 km/h?
10. A car accelerates uniformly from rest at 2.4 m/s^2 .
- (a) What is its velocity after 6.0 s?
- (b) How far does it travel during this time?

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11. A train traveling at 25 m/s applies its brakes and decelerates at 1.5 m/s^2 .

(a) How long does it take for the train to stop?

(b) What distance does the train travel while braking?

12. A ball is thrown vertically upward with an initial velocity of 15 m/s. Use $g = 10 \text{ m/s}^2$.

(a) How long does it take to reach its maximum height?

(b) What is the maximum height reached?

(c) What is the ball's velocity when it returns to the thrower's hand?